

# Faizah Mahendinawaz Kureshi

in [Linkedin](#) | [Github](#) | [✉ kureshifaizah@gmail.com](mailto:kureshifaizah@gmail.com) | [☎ +91937710006](tel:+91937710006)

## EDUCATION

Vellore Institute of Technology (VIT)

Chennai, India

*Bachelor of Technology, Computer Science and Engineering  
with Specialization in Artificial Intelligence and Machine Learning*  
GPA: 3.88/4

2020-2024

## EXPERIENCE

Merai New Age Pvt.Ltd

June 2024 - Present

AI Engineer

- **Robust Explainable AI Model Testing Platform**

- Leading a team of 3 for the development of a **model agnostic Explainable AI Platform** for image-based model
- Developed robust **SHAP** and **LIME** explanations applicable to all **CNN-based** models
- Currently implementing **SHAP** and **LIME** explanations for **transformer-based** models

- **Radioiq.ai** - Lung X-Ray Analyzing tool

- Responsible for developing classification model for tuberculosis, using **CNN-based** model enhanced with an **attention layer** achieving **Specificity** of **81%** and **Sensitivity** of **92%**
- Used **PFT-net** for heart segmentation and lung segmentation via detection to find Cardiothoracic Ratio

- **U Matter** - a mental health app used to predict anxiety and depression

- Analyzed phone's camera feed to detect key indicators for anxiety & depression
- Developed **Custom Transformer model**, used **Open Pose** for fidget detection, pupil dilation, and audio-visual cues
- Used **NLP** methods to analyze psychiatrist-provided assessments, achieving **96% accuracy** overall

- **AI Nurse** - India's first facial-based vital signs monitoring system

- Uses facial video analysis to detect blood pressure, oxygen levels, and heart rate. Built using **computer vision**, **custom transformer**, **RPPG**, and **Custom DL Algorithm**, achieving **94.8% accuracy**

Data Science Intern [\[Certificate\]](#)

Feb 2020–June 2024

- Employed techniques like face detection using **Haar Cascade Classifiers**, **GPU-accelerated operations** for faster preprocessing, and **3D convolutional filters** to reduce temporal noise in the frames to prepare high-quality data for real-time emotion detection with **Residual Masking Networks (RMN)**
- Trained the RMN model on a dataset of 10,000+ Indian user images, leveraging **data augmentation** and **transfer learning** for emotion detection

National University of Singapore [\[Certificate\]](#)

Sept 2023 - Dec 2023

Research Intern

- Developed 'MotionCraft' with Dr. Dash, leveraging fluid hand movements to redefine human-computer interaction, achieving **95% accuracy** in classification with robust gesture prediction using sparse categorical cross-entropy
- Used **OpenCV**, **OpenPose** and **Google MediaPipe**, integrating an **LSTM** for temporal hand tracking and a streamlined **ANN** for keypoint classification, reducing computational overhead by 30%, optimized model training with **ADAM** and **stochastic gradient descent algorithms**, implementing **dropout**, **early stopping**, and **Federated Learning** to reduce overfitting by 25%

National University of Singapore [\[Certificate\]](#)

July 2023

Global Academic Research Intern

- Led the Alertify project to develop an advanced real-time surveillance system, achieving **95% accuracy** in violence detection and automatically alerting authorities via Telegram, including crime location, time, and images
- Utilized **MobileNetV2** with a **custom self-learning rate algorithm** on a benchmark dataset, enhancing model efficiency by 25%. Later improved by using **Deep Sort** tracking and optimized nighttime detection via **Computer Vision**, achieving 30% better low-light accuracy.

# PROJECTS

- AN XAI framework for Complete Understanding of Image Classification Models

Dec 2023 - Mar 2024

- Built a framework to enhance transparency in **CNN models**, showing **98% alignment** between model-predicted features and areas identified by experts. Tested on Brain Tumor Classification and ImageNet datasets for reliable performance.
  - Used **GradCam** to highlight most influential areas in the last convolution layer, **Layerwise Relevance Propagation** (LRP) to identify key pixels that contributed the most in prediction, and **SHAP** to detect misleading pixels in predictions
- Speech-enabled Image Narration: Assistance for Visually Impaired

July 2023 - Nov 2023

- Engineered a system to convert visual inputs from augmented reality glasses into audio output with a **95% accuracy**, using **Google Text-to-Speech**, enabling real-time guidance and enhanced accessibility for visually impaired users
  - Achieved **92% accuracy** in image processing by feeding extracted **InceptionV3** features into an **LSTM** with integrated **GloVe embeddings**, improving caption generation accuracy by 30% for precise and contextually relevant outputs
- Predictive Maintenance for Elevator

Dec 2022 - Apr 2023

- Collected real-time data on environmental conditions, steel rope wear, and brake speed by integrating sensors (**accelerometer, IR, DHT, and sound sensor**) to monitor elevator components, enabling prediction of maintenance dates
  - Developed a predictive maintenance model with **95% accuracy** using **Random Forest model** for service date forecasts, integrated a **Flask server** with **Arduino**, built a real-time frontend app to monitor component health and service needs
  - Achieved a 30% reduction in downtime and improved safety by identifying maintenance needs, preventing failures

# PUBLICATIONS

- Shravan Khunti, Faizah Mahendi Nawaz Kureshi, Rahulkumar Ankola, and Prikshit Kumar. 2024."**Empowering Precision in Financial News: A Revolution in Editorial Classification through Cutting-Edge Natural Language Processing**". In Proceedings of the 2023 7th International Conference on Natural Language Processing and Information Retrieval (NLPIR '23). Association for Computing Machinery, New York, NY, USA, 210–221. <https://doi.org/10.1145/3639233.3639343>
- U. D. P, Faizah Mahendi Nawaz Kureshi, S. Khunti, T. V. Muni, R. Kumar Tata and S. Kalarani, "**Multiauthority Cloud Storage with Secure and Efficient Attribute-Based Access Control**", 2023 Global Conference on Information Technologies and Communications (GCITC), Bangalore, India, 2023, pp. 1-4, doi: 10.1109/GC-ITC60406.2023.10426217
- Presented "**AI-Driven Predictive Models for Early Chronic Disease Detection & Personalized Prevention Strategies**" at the International Conference on Smart Cyber Physical Systems to be published in Springer Journal
- "**Explainable Deep Learning for Time-Series Medical Data: Enhancing Interpretability in Clinical Diagnostics**" has been accepted in International Conference on Information Technology and Artificial Intelligence 2025 and is to be published in SCOPUS Indexed Springer Book Series, 'Lecture Notes in Networks and Systems'

# TECHNICAL SKILLS

- Programming:** Python(advanced), C/C++, Java, SQL
- ML/AI:** TensorFlow, PyTorch, Scikit-learn, Generative AI, NLP, Computer Vision, LLM, RAG, Transformer Architectures, Explainable AI (XAI), Reinforcement Learning (RL)
- Tools:** Docker, Kubernetes, MATLAB, Git/GitHub